This is an excerpt from Frank Elavsky's dissertation on *Tool-making as an Intervention on the Accessibility of Interactive Data Experiences*, which can be accessed in full at this archival link (once published):

http://reports-archive.adm.cs.cmu.edu/anon/hcii/CMU-HCII-26-103.pdf

**This document contains the following sections:**

- Chapter 1: What do we mean by "accessibility" and "tools?"
- Chapter 9: Findings (Sections 9.2, 9.4, and 9.5 only)
- Chapter 10: Discussion & Future Work (Sections 10.1 and 10.3 only)
- References

**This document does not contain the following chapters:**

- Chapter 2: Background & Related Work
- Chapter 3: Overview of Contributions
- Chapter 4: *Chartability*: Heuristics as a Tool and Resource
- Chapter 5: *Data Navigator*: Low-level Tooling for Creating Navigable Data Structures
- Chapter 6: *Skeleton*: Visual Authoring of Non-visual Data Experiences
- Chapter 7: *Cross-perception*: Rethinking Input Design Towards Blind Analytical Interaction
- Chapter 8: *Softerware*: Enabling Personalization of Interactive Data Representations for Users with Disabilities
- Chapter 12: The *Softerware* Option Taxonomy
- Chapter 11: Biographical Sketch

# Part I

# Introduction

# Chapter 1
# What do we mean by "accessibility" and "tools?"

This thesis is a body of research situated within the existing research area focused on making interactive data representations more accessible for people with disabilities. Much of the work in this existing area is situated within the context of making interactive data *visualizations* accessible, particularly (but not exclusively) for people who are blind. Our work, contributed here in this thesis, is focused on using *tools* as a specific intervention and sub-area of study for making interactive data *representations* accessible for people with disabilities, broadly speaking. ("Representations" here is an intentionally broader term than "visualizations," which are exclusively visual representations of data.)

Before we begin, two things must be understood up front, or else the rest of this thesis could be interpreted with disruptive assumptions: we must interrogate the phrase "making visualizations accessible" and unpack why *tools* are a meaningful area of study.

## 1.1 Is "accessible visualization" really an oxymoron?

The first assumption that must be disrupted is perhaps the motivating cornerstone of this research, which is that the phrase "making visualizations accessible," while a noble goal, is not the semantically correct phrasing nor precisely what describes our work. This can be misleading. We do use the phrase "accessible visualization" but will admit that this seems to confuse certain people with very particular opinions about things. We will clear this up.

Villains of our field's past have written incendiary and ableist perspectives on why "no forms of data visualization, not just dashboards jam-packed with graphics, can be made fully accessible to someone who is blind," and that "[a blind man] will never be able to analyze data as I do visually, because many aspects of vision cannot be duplicated by his other senses" [8]. However, this position misunderstands what the goal of accessibility is, and arguably even what the goal of visualization itself is.

Making visualizations accessible *isn't* about the visualization, it's about making the outcomes of the visualization accessible.

Visualizations are ubiquitous and paramount for decision-making. However, the *artifact* that is a visualization is not even the goal of the act of visualizing: developing understanding, insight, confidence, and communication among and between human beings are the goals of visualization. Visualization is about making data easier to use for all kinds of things. Yes, our visual system enables us more than any other form of sensory cognition that we have [5, 10, 31]. But we aren't trying to make sight itself accessible. We are trying to make it possible for people to make meaningful decisions, gain valuable information, build conjectures, and effectively communicate with others.

Many, many people who I, Frank, have spoken to over the course of my career, even before embarking on this thesis journey, misunderstand this simple fact: making a "visualization" accessible *isn't* about the visualization itself but rather making what the visualization is meant to *accomplish* accessible. It's about equal outcomes, not equal interactions with an artifact.

People with disabilities are no small portion of the world's population. In the United States, 26% of people self-report living with at least one disability that affects their daily lives [23] and all of us will eventually age into disability (if we are lucky to live a long life).

People with disabilities deserve to participate fully in life. They deserve financial independence. They also deserve loving care and interdependence [2, 16, 21]. People with disabilities have a right to make informed decisions, to know about the status of a global pandemic, and to have an understanding of local and national politics [7]. While we use visualizations to navigate all of these domains, the goal is not to make the charts and graphs themselves somehow equally useful to all people. That would be a false measurement of success.

*Our goal then, is measured by the success of lives led by people with disabilities* [32]. Many other measurements are just metrics along the journey towards that goal. We then ask: Can people with disabilities also use data to live full lives? Can they make *fast* decisions based on data? Meaningful, careful, *slow* decisions? Communicate complex ideas? Crunch and clean data, develop models, find errors, and build hypotheses? Can they have memorable, immersive, beautiful, aesthetic experiences with data too [16]? Ultimately, our aim is not accessible "visualizations" but equitable and accessible *outcomes*, everything an interactive data experience *accomplishes* for the people who use it.

Again, if the goal of accessible visualization were about visualizations themselves, then the correct course of action would be one framed by the medical model of disability [20]: that there is a normative state of behavior and capability (in this case, it would be "normal" to be able to read a visualization and make a decision) and any deviation from that norm must be corrected. This framing first assumes that the visualization should not be altered or improved. And then this framing puts the burden on the bodies of people with disabilities: that they must be "fixed" and given sight or brought to some equivalent state as someone who is "healthy," normal, and sighted. Plenty of scholars have already discussed why this framing is a problem, not only because it places undue burden on people with disabilities and produces pathologies and hierarchies of disability, but also because it is fundamentally not economically or ethically feasible.

So we then turn to other models of disability, such as the social model. The social model is heavily discussed by disability scholars and is not the end-game or last and total way of thinking about disability [20, 24, 25, 30, 40]. But the core motivation is that society, not medicine, is also a path towards solving problems that people with disabilities face. A few important concepts and concretely actionable things come from the social model that can help motivate the work of this thesis.

First, we look to the historical birth of the social model of disability: in the 504 sit-ins that took place in the United States in 1977. Cities had curbs and curbs are a barrier for people who use wheelchairs. So protests happened because decisions were being made without people with disabilities at the table. In this instance, people acknowledged that political power was an exclusive club and fought to ensure their cry "nothing about us, without us!" materialized.

And this leads us to the first and most-foundational philosophical framing for this thesis: that our *artifacts*, these things we've created from curbs to data visualizations, can become *barriers* for people with disabilities. And it is then the artifact, not the body of the person with a disability, where disability is produced in this model. Rather than a comparison to a normative state as a way to frame disability (the medical model), we instead must observe and evaluate material outcomes based on human-made problems.

So, the social model is framed around society "solving" inequities: we get involved and make political and legal change tangible. But a second model also emerges from within the social model: one where we can now frame *who is first responsible* for repair: the curb designers and implementers.

And knowing who is first responsible for access leads us into the moral and ethical imperative that motivates this thesis: the builders and makers of visualizations are ultimately the ones who provide exclusive value for only a subset of people: those *without* disabilities. **We must first change how builders and makers do their work.**

So the phrase "accessible visualization" is really about recognizing that visualizations produce barriers for people. That means that it is our ethical imperative, as builders and makers, to fix them. And that act of fixing barriers leads us away from mere visual representations of data into a wide variety of other senses and interaction modalities. There are many paths forward towards fuller and more-equitable lives led by people with disabilities.

## 1.2 On *tools*, *tool-making*, and *human-tool* interaction

Then the act of making becomes immensely important: we, the builders and makers of our world, need to get things right; there is a risk involved when making things that we will exclude people with disabilities. We need to make sure that we build a better world than the one we have now. We must care for new things we create and tend to the repair and maintenance of what we've already made. And this ethical imperative leads us to the topic of *tools* and *tool-making*.

So the second thing that must be understood before we embark on this thesis is that *tools* are not the same as *solutions* or *applications*. Sometimes tools can be used to *solve* things and are certainly, in ideal circumstances, *applied* in various contexts. But understanding the role of the "tool" in human-tool interaction is paramount for engaging in the work of making anything accessible for people with disabilities.

We use tools to shape our world, break old things, and make new things. But a tool, like the hammer (as an example), does not inherently *solve* something like homelessness. But a hammer can be used to build homes if there are social policies in place and proper resources allocated. This means that for the success of tooling, there is often a larger material, social, legal, and policy reality that supports and necessitates those tools. This thesis will not be focusing on changing the upstream dependencies, but optimistically operating as if they were true (or will be true in time).

However, in some cases, tools can *destroy*. The hammer has a claw and can easily pry apart boards and tear down homes. So tools carry potential to do all kinds of things, both good and bad, and how a tool is used is often open-ended, variable, and heavily dependent on socio-technical realities. Tools participate in personal and political agendas [37] and are sometimes, for this reason, regulated or made proprietary and controlled by powerful entities [12, 35].

So tools are not without any sort of ethics. We cannot just blame tool-users for outcomes when much of a tool depends on these larger systems and structures. Technologies (tools included) encode the assumptions and biases of their *creators* as much as, if not more than, their users. Tools that build things for others to use can be loaded with assumptions about what people are *able* to do [38] and also rules and guardrails about what anyone downstream from that tool's design *should* do [12, 34]. These assumptions, biases, and rules *limit*, *enforce*, *magnify*, *exclude*, and *enable* what a tool-user is capable of.

Tools for visualizing data are a perfect case study in this problem: virtually every major data visualization library, application, or software ever made was made entirely with the assumption that data should be transformed into visual representations. This is a reasonable assumption, since virtually all of the tool-makers are sighted and visualization is incredibly helpful to our cognition of and communication with data [9].

So data visualization, as a field, has focused its tool-making efforts on reducing the difficulty involved in visualizing data. Some visualization tools are concise [28], others are lower level but much more expressive [4]. Tool-making in visualization has focused on making it easier to scaffold a wide variety of interactions both with the visualizations as well as with their underlying data models [14].

However, as time has moved on, people began to speak out about color-vision deficiency in data visualization. Some people, primarily those with X/Y chromosomes (largely men) who are of European ancestry, have a deficiency in their ability to perceive certain colors. Then a plethora of research arose that began to look into the barriers that folks who are colorblind face in data visualization. As a result, our practices and tools improved. We began to educate practitioners, develop new color palettes, researched new methods for testing our designs, and built new systems for handling automatic color encoding. Our tools evolved.

But now data visualizations have arguably become ubiquitous in daily life. By comparison, we have far more tools now for making visualizations quickly and easily than we do for representing data in non-visual ways. We also have far more research, relatively speaking, into how sighted end users interact with visualizations.

So this thesis engages gaps that arise in this space: Practitioners face immense challenges when crafting accessible data experiences. We first need to educate practitioners on what accessibility barriers actually are in interactive visualizations. Then, we must help them engage the hardest barriers in this work and create building blocks that help them to construct navigable data experiences, build design frameworks that can inform entirely new kinds of data interaction, and develop software systems for end-user personalization and agency. Our research seeks to advance the state of the art in tools that assist in accessible data interaction while also using tool-making as an intervention that helps us to better understand and characterize *why* and *how* data practitioners face barriers themselves in this work.

# Part VI

# Conclusion

# Chapter 9
# Findings

The five projects in this thesis were presented as five chapters, each with its own results and its own conclusion. Read separately, they contribute a heuristic resource for evaluation, a structural toolkit for navigation, an authoring environment for non-visual design, an interaction technique and device for blind analysis, and a system and study of end-user personalization. This chapter states what they show *together*. Chapter 2 closed on a lopsided field: a breadth of empirical work and exemplary prototypes on one side, and a near absence of practitioner tools in production environments and practitioner action on the other, with the hardest remaining challenges concentrated in navigation, interaction, and personalization. The findings below are what this thesis learned by working inside that gap.

Each finding is a synthesis across chapters, and each is traceable to results reported in the body of this thesis. We have written them for two audiences at once: practitioners who build, evaluate, and maintain data experiences, and researchers who study how that work happens. Where a promising idea outran the evidence, it is not here; it is in the next chapter, with the rest of the discussion and future work.

## 9.2 Tools Can Reduce the Work Required to Make Things Accessible

Chapter 2 argued that guidance cannot substitute for means. The constructive half of that argument is that supplying means measurably changes what practitioners can do, and three chapters provide the evidence.

*Chartability* reduced the perceived difficulty of evaluation and increased the confidence of practitioners new to accessibility, whose audit results were reasonably comparable to the authors' own; beyond the study, it has been applied by policy and governmental organizations and more than 100 companies, across toolchains from Tableau and Excel to D3 and Figma. *Data Navigator* gave previously inaccessible rendering formats a navigable structure: a static raster image gained full multi-input navigation with no automatically detectable conformance failures, and a deterministic ingestion pipeline recreated, and then improved on, an existing toolkit's navigation scheme for rendering modes that previously had none. *Skeleton*'s scaffolding collapsed the spatial authoring of a navigation structure to one or two minutes of refinement, and within single sessions every participant iterated substantively on designs that, in a code-only workflow, our co-design collaborators had struggled to iterate on at all.

None of this is evidence that tools solve access; the boundaries of that claim get their own finding below. It is evidence of something narrower and more useful: the labor of access work is sensitive to tooling, and reductions in that labor are achievable, measurable, and repeatable across very different kinds of tools, from a workbook of heuristics to a rendering-independent toolkit to an authoring interface.

## 9.4 Friction From a Tool Can Be Productive

Perhaps in tension with the previous finding that tooling can reduce work, tooling may also *increase* some kinds of work in exchange for quality. The assumption for tooling, in a classic or naïve sense, is that tools exist to make work easier by reducing the labor and friction of certain tasks. But ease of work is only a goal if a baseline exists. If some work isn't being done or new design possibilities open up, then *more*, not less, work might be the goal of an idealized tool. Friction can be useful, even productive, and sometimes necessary. While outside of the scope of this thesis, we would urge fellow researchers to explore the tensions that arise in design when removing friction is the primary, if only, goal of a tool (as it appears to be the case with modern large language models and agentic tooling).

But it *is* a reasonable assumption about practitioner tooling that its job is to make work smoother and easier. The evidence in this thesis complicates that assumption: in several places, the value of a tool came precisely from the resistance it introduced. *Chartability* forced engagement with conflicts between guidelines (such as redundant encodings that help some users while overwhelming others) rather than smoothing those conflicts away; audits remained slow, and participants reported that the slowness helped them focus and prioritize. Evaluating a visualization for accessibility actually *adds* time to the overall time it takes to make a visualization. *Chartability* made it clear that if work isn't being done at all in the first place, introducing new heuristics to a practitioner's workflow ultimately generates friction that wasn't there before. We view this as necessary and good.

*Data Navigator* arguably introduced so much additional work and friction that it necessitated additional tools to make it easier to work with (in the *Inspector*, *Dimensions API*, and ultimately *Skeleton* of Chapter 6). It became apparent that giving practitioners a nearly-infinite new space of design without clear scoping and bounds for that design space was burdensome and resulted in many additional barriers and design iterations. *Data Navigator* could create *any* discrete navigation experience from relationships in an information space. This is fantastic. But it appears that *scope* in design is, in its own way, a source of friction *and* a relief on the work required of that design: it limits and holds back what can be done. But this scoping friction in turn reduces the labor of exploration required. Infinite, frictionless exploration of a design space appears to have put friction elsewhere in the work: during deliberation, handoffs, testing, and ultimately in producing scope.

*Skeleton*'s sessions opened on an intentionally problematic default structure, and its testing stage made designs that looked adequate feel inadequate in traversal; participants responded to that resistance not by abandoning the work but by iterating, restructuring dimensions, and in one case restarting from the wizard. The tool introduced *more*, not fewer design iterations. Interestingly, *Skeleton* was intended to reduce friction and make work easier, yet by surfacing and making perceivable and manipulable a space of design elements, practitioners didn't just neatly solve problems in less time or solve more problems, in some cases they problematized their own work and returned to the figurative (and literal) drawing board.

*Softerware* also produced friction at the level of professional identity: it challenges the traditional separation between the people who design a system and the people who use it, and our practitioners immediately raised the ethical concern that personalization could excuse poor defaults. That concern surfacing early, before any deployment, is the friction doing its work.

Practitioners suddenly had to worry about ways that users could self-design into misleading, harmful, or newly inaccessible designs on their own. And yet, practitioners then had to wrestle with the question, "if we give people power over their interfaces, isn't it then their responsibility in the end for what they do with that power?" Our prototype and study didn't neatly resolve these frictions or close out this conversation, but rather these frictions simply served to open up new directions and outline problems presently unsolved.

We would argue that the *cross-feelter* device from *Cross-perception* was a "classic" tool in this sense, unlike the rest: it reduced friction *and* reduced work, which led to more productivity. It is good to hold this project in tension with all of the others in this thesis, which introduced new, productive frictions into accessibility work in some way or another and may or may not have made work easier in some ways in the process. *Cross-perception*, by comparison, primarily served to open up a space of design while we demonstrated one design within that space through our hardware device.

Chapter 2 argued that tools direct the work and the workers who use them. Friction is one of the ways they do it. The finding here is that a tool which removes all resistance from accessibility work can also remove the engagement that the work requires. This is, in some cases, not an ideal outcome. Engagement with particular tensions might be the *best* thing that a particular tool can accomplish. This means that toolmakers can place friction deliberately: at conflicts that deserve a decision, at defaults that deserve suspicion, at moments where compliance would otherwise substitute for judgment, and in places where designerly, considerate work isn't being done but *should* be.

## 9.5 Tool-use and Tool-making Are Generative

Across the studies in this thesis, tools did not merely execute their users' existing intentions; they expanded what practitioners and users could imagine. In *Skeleton*'s study, five of eight participants reconsidered the design of the visualizations they had brought, not just the navigation built over them: some simplified, some questioned their chart type, and one concluded that the right non-visual design was no navigation at all, a genuine design decision rather than an omission. In *Cross-perception*, formalizing the technique and putting a working device in participants' hands opened an input design space: blind data practitioners proposed pre-rendered tactile interaction cubes, feelter-driven array navigation, sonification scrubbing, and multi-axis spatial interaction, extensions we present as design starting points. In *Softerware*, the visible option space provoked interface ideas we had not designed for, including in-context, voice-driven adjustment while navigating, and led practitioners to imagine authoring their own systems through a softerware interface. Even *Data Navigator*, the lowest-level contribution here, showed this property (and perhaps the strongest): its vocabulary of nodes, edges, and rules gave co-design sessions a shared language in which novel navigable prototypes could be drafted in days.

The generative freedom afforded by tools isn't just *expressiveness*, as defined by existing work on visualization grammars [28, 36], which tends to enable fluency within the bounds of a structured language, but also generation outside of any existing bounds as well [1].

For researchers, this finding doubles as a methodological one, and it is the stance Chapter 2 named: studying practitioners through their tools works because a tool placed in real work is simultaneously an intervention and an instrument. The generative responses it provokes, the reconsiderations, extensions, and refusals, are data about why the work is hard and where it could go next.

And tool-making isn't just generative because tool-users appropriate, re-use, and innovate once they have a new tool, but viewing gaps and problems that practitioners face is also generative for new research directions. This was the cornerstone motivation behind our *action research* approach in Chapter 6: that collaboration with practitioners as an embedded designer, engineer, and researcher in order to join them as they solve problems provides a rich space where novel intellectual gaps can be made both legible and actionable. Tool-making alongside tool-makers and tool-users without having an *a priori* intellectual agenda in mind leaves room for developing new conjectures, evaluating multiple collaborations in aggregate, and ultimately generating research agendas.

# Chapter 10
# Discussion & Future Work

## 10.1 What is a "tool?" A reflection on the social and material identity of tools

In the introduction of this dissertation, we use the example of a hammer: a hammer can destroy and it can construct. So is the *use* of a technology what constitutes it? Do we understand the hammer as the *thing we swing, to destroy and to build*? Should we?

This thesis engages domains of tools and tool-making for accessibility: evaluation, navigation, interaction, and personalization. But these categories for work do not fully characterize the upstream conditions that our software systems and data interfaces inherit.

In our work specifically on accessibility, a larger social reality becomes apparent that shapes the question, "what is a tool?" far more than how an individual might use one, or the domains of work that our tool-making inhabits. Our research has navigated multiple social and political thresholds, from changes to law in the European Union, to the enactment of Title II as part of the update to the Americans with Disabilities Act. These laws have motivated a significant interest in accessibility research, solutions, guidelines, and technologies. In the midst of this, we have seen the rise of overlays and generative AI solutionism [13] and subsequent lawsuits and grass-roots resistance.

For our work, this is mostly good news. Legal change produces motivation, and even with predatory technology attempting to address real problems, pushback is widespread and active. But this paints a picture of the reality that our work inherits: many tools cannot even be used, or cease to be used, if there is not a social, political, and material set of conditions in place motivating those tools, providing resources for their construction, regulating their use, and examining the outcomes of what they accomplish. Tools and technologies are often a response to social, cultural, political, and legal realities that we first negotiate.

I, Frank, recently spoke on this at a keynote in Australia, on how a hammer isn't *just* a tool and that the idea that "the only thing that matters is how a tool is used" limits how we really understand tools. Instead, I spoke about how a standard, household hammer requires iron and wood. That alone leads to a whole universe of different questions. Western Australia's conservation efforts were disrupted when a significant amount of natural iron was discovered in a wildlife preserve. So laws were passed and now iron is mined there. That iron is largely exported. And Australia then, whether with Australian iron or not, mostly imports their small tools. Iron is sent out, and through a complex network of trade (likely indirectly related to the iron), hammers are brought in. A "hammer," to even exist at all, relies on multiple levels of human governance, international relations, and complex infrastructures of trade.

And while my metaphor is largely motivated to encourage younger practitioners to consider the "iron mines" in the technologies they use, such as modern generative AI, it is also an area that is not adequately explored and addressed in terms of accessibility research.

Research on accessibility is dependent on funding, which is often dependent on political priorities and action. Depending on the current social and political state of the world at large, accessibility research itself may never gain the opportunities required in order to innovate and produce new tools at all. And as the US's 2025 federal cuts to research demonstrated, millions of dollars devoted to accessibility research can be lost to political agendas. It is for this reason that engagement with policy recommendation and guidance is essential. Personal political activity and involvement is also essential. Researchers who genuinely believe in accessibility as a human right or as a dignity that all people deserve should work with policymakers to ensure that there are material and structural resources in place for this work to continue. We cannot naively believe that technology, divorced from the realm of social and political forces, is capable of solving accessibility barriers [30]. Without enforcement and threat of litigation, very little accessibility work has been accomplished in the past by technology companies alone.

Featured in these chapters (in their pre and postfaces) is the policy and outreach work involved in seeing that work like *Chartability* and *Data Navigator* were used in real contexts, including by organizations that govern and influence the lives of many people. Immediate incentives to produce novelty may not be enough to sustain the larger socio-cultural and political ecosystems that our work participates in and is downstream from. We must also get involved.

## 10.3 Who is responsible for repair?

Lastly, we want to revisit one of this thesis's opening points, where we argue that the *tool-makers* are first responsible for repair. This is true. However, the most pressing issue we have faced in recent years is mostly unmentioned across these research projects: tool-makers might be responsible, but this is because they are the only ones who have the *power* to make things accessible. Does this always need to be the case? Can we imagine an artifact's authority over the user's ability to access being designed towards self-subversion [11] or de-centralized agency [6, 18, 22], instead? What might that look like, concretely?

In *Softerware*, we begin to engage this larger problem in terms of an idealized state where a user can repair or re-design their own experiences. But to me, this self-repair is like laying down train tracks for yourself as you move a locomotive, but then lifting up your own tracks behind you as you go. You're the only one helping yourself. This is not ideal, for you or others.

What we need are broad, lasting, infrastructural changes. On the web, this problem becomes quite difficult to solve. A personal computer or device? Again, someone can auto-design their interfaces into a better state. But back when we started *Chartability*, the WebAIM Million's report showed more than 95% of the top one million website home pages contain at least one critical accessibility error. And now, more than six years later, that proportion is unchanged [33].

Some had imagined that generative AI would solve the massive infrastructural repair problems we face. But unfortunately, the latest WebAIM Million report shows that since 2020, ARIA usage has increased and correlates to more errors, while use of tabindex on a page has increased nearly 300% and also correlates to more errors on a page. If anything, during the age of generative AI, we have seen existing bad patterns worsen in prevalence and complexity.

We firmly believe that a tools-based approach is not enough on its own. Tool-making cannot be the *only* intervention on inaccessibility. Tools and tool-making, as our thesis argues, have a powerful role to play. But we simply can't tool our way out of failed infrastructure and inadequate policy when someone else *owns* the tools and tool-making. Visiting a website is like going into someone else's home: arranged according to their effort, tastes, and so on. If you can't access their home, you essentially need to request that they let you in personally. Website repair always falls to the owner and maintainer of a website, and they largely don't take any meaningful action.

Sidewalks outside of homes are a good parallel to this problem. Sidewalk accessibility is a massive infrastructural problem [27], and yet localities treat sidewalk maintenance in different ways: some, like where I, Frank, presently live in the south hills of Pittsburgh, put the onus on the homeowner whose house and property the sidewalk touches. In other places, sidewalks are considered a public path, similar to a roadway, and are maintained through public tax and resource management. To no surprise, privately-maintained, public-access sidewalks are worse for people in pretty much every way than publicly-maintained ones [39]. This is because private homeowners don't care about sidewalk maintenance unless the city manages to fine them or they get sued.

And the web is a collection of private spaces that you visit privately. There is no truly shared, universally democratic, public space on the web. Centralization is partly to blame: sharing space while scaling leads to consolidation.

So our future work will continue to wrestle with the same tensions of scale, repair, and anti-consolidation of power, motivated by the same WebAIM Million report. But now we look to questions of *democratic* and *radical* access to accessibility repair. The barriers we hope to tackle in the future are political and infrastructural. Perhaps tool-making will participate in this work, but it seems clear now from our work that the upstream technical problems and socio-political conditions that tools inherit, will likely not be addressed by tools alone.

What does "democratic" and "radical" infrastructure work look like? It will probably be an extension of *Softerware*, to some degree. We imagine future research that explores public-first spaces, ones where access is socially negotiated and repair belongs to all of us. Is this an autonomous space, like an autonomous zone [3] separate from the web? Above it, looking down into it, like shared annotation tools but capable of sharing the manipulation of websites [26]? A space with ambient co-repair, modeled after projects that bring people together [29] or that allow community "fixing" of misinformation [17]? Perhaps feminist thought on the ethics of care can help us [15, 19]? Or maybe it will be something else entirely; I'm not yet sure. But what made the web fantastic years ago is long gone; most of it has been hedged into corporate spaces that are controlled, maintained, and repaired by corporate power. And these entities are notoriously bad at repair. What we imagine in the future involves reclaiming a sense that the web is *ours*, belongs to *us*, and that ultimately *we* are responsible for making it accessible.

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